

Supplementary Information

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Mechanistic simulation of heritable expression states, cell division and drug response in single-cell populations. Supplementary Notes 1–10, Supplementary Figures 1–2 and Supplementary Tables 1–4.

Supplementary Note 1: Stationary laws used for validation

Birth-death. For $\emptyset \xrightarrow{k} X \xrightarrow{\gamma} \emptyset$ the stationary distribution is Poisson with mean k/γ .

Telegraph model. With activation k_{on} , inactivation k_{off} , transcription k_{tx} from the active state and degradation γ , and $a = k_{\text{on}}/\gamma$, $b = k_{\text{off}}/\gamma$, $\lambda = k_{\text{tx}}/\gamma$, the stationary distribution is the Beta-Poisson mixture

$$P(n) = \frac{\lambda^n \Gamma(a+n) \Gamma(a+b)}{n! \Gamma(a+b+n) \Gamma(a)} {}_1F_1(a+n; a+b+n; -\lambda),$$

whose mean is $\lambda a/(a+b)$ and whose variance is $\mu + \lambda^2 ab/[(a+b)^2(a+b+1)]$. We evaluate ${}_1F_1(\alpha; \beta; -\lambda)$ through Kummer’s transformation $e^{-\lambda} {}_1F_1(\beta - \alpha; \beta; \lambda)$, whose series has positive terms; partial sums are rescaled by 10^{-200} whenever they exceed 10^{200} so that the log of the sum is accumulated without overflow.

Bursty protein. Protein produced in geometric bursts of mean size b arriving at rate a and degraded at rate γ is negative binomial with shape a/γ and success probability $1/(1+b)$. The package emulates bursts through an mRNA of lifetime $1/(200\gamma)$ translated at rate $200\gamma b$.

Supplementary Note 2: Kernel details and correctness tests

The direct method draws the waiting time from $\text{Exp}(a_0)$ and the channel by linear search over the cumulative propensities; after each event only the propensities of reactions that depend on a changed species (the dependency graph) are recomputed, and the running total is refreshed every 512 events. Fixed-step tau-leaping draws $\text{Poisson}(a_j \tau)$ firings for every channel; the strict variant errors when a species would become negative, the hybrid variant rejects the leap and simulates the interval with the direct method. Adaptive tau-leaping classifies as critical every channel within $n_c = 10$ firings of exhausting a reactant, selects τ from the non-critical channels through the bounds $\max(\varepsilon x_i/g_i, 1)/|\mu_i|$ and $\max(\varepsilon x_i/g_i, 1)^2/\sigma_i^2$, draws the time to the next critical event from $\text{Exp}(a_0^{\text{crit}})$, fires at most one critical channel, halves τ on rejection, and performs 100 exact events whenever $\tau < 10/a_0$.

Supplementary Table 1 lists the tests of the package test suite that validate the kernels: Kolmogorov-Smirnov distances below 0.015 (direct) or 0.02 (approximate kernels) to the Poisson, Beta-Poisson and negative-binomial laws with 8 000 to 12 000 cells; agreement of the adaptive kernel with the direct method on a stiff dimerisation network within 5% in the mean; identical results with one and with several threads.

29 Supplementary Note 3: Parameters of all simulations

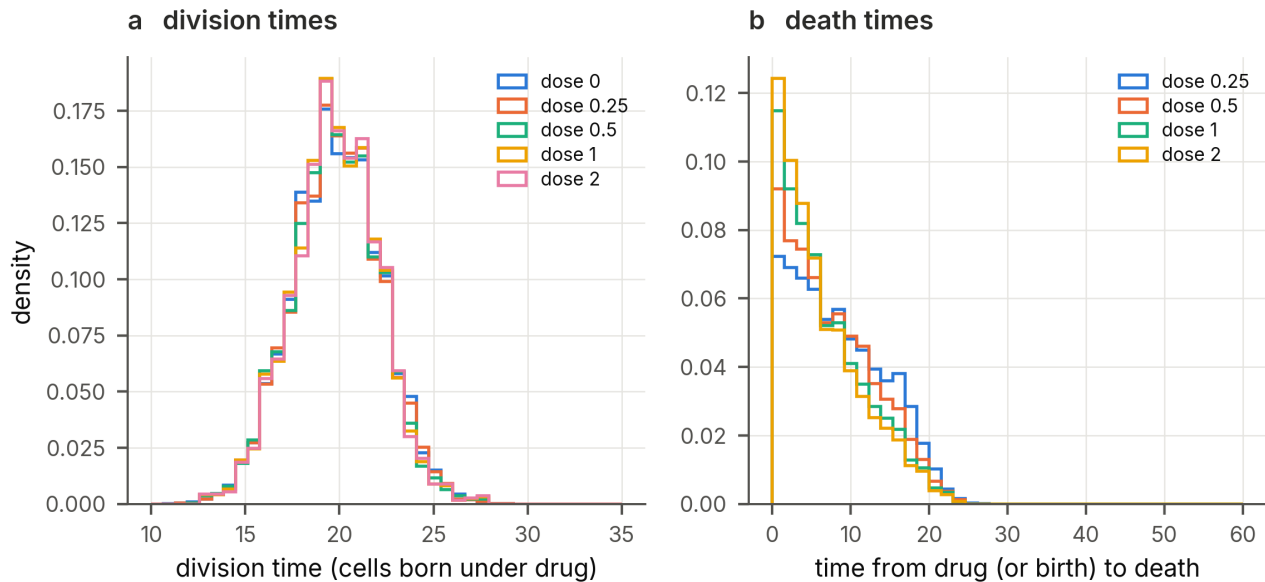
30 Supplementary Table 2. Parameters of every simulation in this Article.

Figure	Model	Parameters
2a	birth-death	$k = 12, \gamma = 1, T = 15$
2b	telegraph	$(k_{\text{on}}, k_{\text{off}}, k_{\text{tx}}) = (0.4, 0.6, 12)$ and $(0.05, 0.5, 40), \gamma = 1, T = 40$
2c	bursty protein	$a = 1.5, b = 6, \gamma = 1, T = 25$
2d-e	telegraph	$(0.4, 0.6, 12), n = 20\,000$
2f	random telegraph GRNs	$n_{\text{act}} = n_{\text{genes}}, n_{\text{inh}} = n_{\text{genes}}/2, \lambda = \ln 2/20, dt = 0.1, T = 20$
3	telegraph gene with protein	$k_{\text{tx}} = 30, k_{\text{dm}} = 1, k_{\text{tl}} = 4, k_{\text{dp}} = 0.2, k_{\text{on}} = k_{\text{off}}$ as indicated; $\lambda = \ln 2/20$, sizer $V_{\text{div}} = 2$ (cv 0.05), $\sigma_f = 0.02$, replication at 50% of the cycle where indicated
3e	stable protein	$k_{\text{on}} = k_{\text{off}} = 5, k_{\text{tx}} = 20, k_{\text{dm}} = 1, k_{\text{tl}} = 2, k_{\text{dp}} = 0.05$
4	resistance gene	as Fig. 3 with $(k_{\text{on}}, k_{\text{off}}) = (0.002, 0.02)$ (memory, $p_{\text{on}} \approx 9\%$) or $(0.05, 0.5)$ (fast, same p_{on}); founders burnt in for 300 time units at constant size; death $h_{\text{max}} = 0.15, \text{EC}_{50} = 0.5, m = 2, K = 150, q = 4$; drug from $t = 40$ of the treatment stage; drug-induced variant $(0.0002, 0.02)$ with $k_{\text{on}} \rightarrow k_{\text{on}}(1 + 100d)$; fitness cost variant: growth $\times (1 - 0.5 c_P^4 / (150^4 + c_P^4))$; memory disruption: switching $\times 20$ on $[0, 40)$ or $[0, 120)$; schedule scan: 300 founders, 240 time units, 20-time-unit exposures
5a	40 independent telegraph genes	$k_{\text{on}} = k_{\text{off}} \in [10^{-3}, 1]$ (log-spaced), $k_{\text{tx}} = 20, k_{\text{dm}} = 0.5$; 40 founders, 6 doublings
5b-c	random GRN	30 genes, 30 activations, 15 inhibitions, $k_{\text{tx}} \in [5, 30], K \in [2, 10], n = 2$, basal 0.05; 1 500 cells; sequencing capture 0.15 (cv 0.3)
5d	random GRN	20 genes, 22 activations, 10 inhibitions; knockdown factor 0.05; 800 cells

Figure	Model	Parameters
6a-b	telegraph	truth (0.3, 0.6, 20); priors log-uniform on [0.01, 10], [0.01, 10], [1, 200]; 200 (a) / 120 (b) particles
6c	resistance gene	truth $h_{\max} = 0.5$, $K = 150$; priors log-uniform on [0.05, 5] and [20, 1000]; 100 particles
7	resistance gene, hours	cell cycle 24 h (sizer, cv 0.1), $k_{\text{tx}} = 30$, $k_{\text{dm}} = 1$, $k_{\text{tl}} = 4$, $k_{\text{dp}} = 0.2$ per h, $K = 150$, $q = 4$, growth-arrest Hill coefficient 2; fitted: $k_{\text{on}}, k_{\text{off}} \in [10^{-3}, 0.3]$, $h_{\max} \in [0.005, 0.5]$ per h, $\text{EC}_{50} \in [3, 40]$ μM , $m \in [1, 6]$, $\text{IC}_{50} \in [1, 40]$ μM ; 300 founders burnt in for 10 cycles, 48 h drug-free, 72 h drug; 160 Latin-hypercube points and 80 Nelder-Mead iterations with seed 1
8a-d	melanoma-like	$k_{\text{tx}} = 3$, $k_{\text{dm}} = 0.1$, $k_{\text{tl}} = 0.4$, $k_{\text{dp}} = 0.02$ per h; net doubling 4 weeks, memory 5 generations, pre-resistant fraction 0.005, $h_{\max} = 0.0023$ per h, $\text{EC}_{50} = 0.3$, $m = 2$, protection $K = 150$, $q = 4$, growth arrest $\text{IC}_{50} = 0.3$ with the same protection, $k_{\text{off}} \rightarrow k_{\text{off}}/(1 + 9d)$, fitness cost 0.5, partial protection: additional unprotected growth inhibition with $\text{IC}_{50} = 1$; 2 000 founders with promoter states drawn from the stationary distribution, 60 weeks, dt 4 h; progression at $1.73 \times$ nadir after the 8-week lead-in

Figure	Model	Parameters
8e-g	MGMT model	same expression kinetics; net doubling 40 days, memory 4 generations, MGMT-expressing fraction 0.01 or 0.30, $h_{\max} = 0.03$ per h at peak, $EC_{50} = 0.4$ of the standard bolus peak, $m = 2$, $K = 150$, $q = 4$; elimination half-life 2.1 h; MGMT consumption $k_{dp} \rightarrow k_{dp}(1 + 20d)$; 1 500 founders at the stationary promoter distribution, six 28-day cycles, dt 1 h

Supplementary Note 4: Additional persister simulations



Supplementary Fig. 1 | Single-cell timing distributions underlying Fig. 4c. **a**, Division times of cells born under drug and **b**, times from drug start (or from birth, for cells born under drug) to death, per dose. The division-time distribution is set by the sizer and is the same at every dose, because in this model the drug kills but does not slow growth. Times to death are broad (coefficient of variation 0.72 at dose 0.5 and 0.82 at dose 2) and their mean shifts by much less than the population decay rate, because most deaths occur among low-expressing cells whose hazard is already close to $h_{\max}$ at dose 0.5 ($EC_{50} = 0.5$, $m = 2$); raising the dose mainly shortens the survival of cells with intermediate protection. The dose dependence of population decay is therefore carried by the fraction of cells that are protected, not by the kinetics of death of unprotected cells, which is the interpretation¹ give of their single-cell tracking data. Under continuous dosing at the highest dose (release period 0, dose 2 in Fig. 4f), the long-term growth rate is

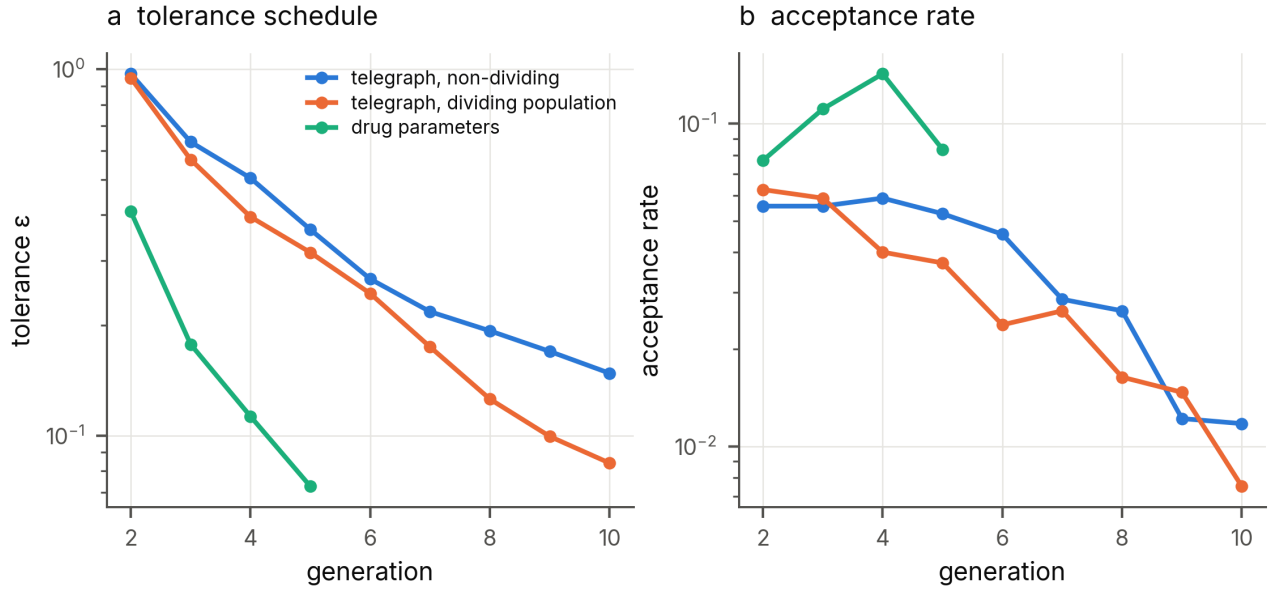
43 +0.004 per time unit for pre-existing tolerance, -0.013 when the resistant state carries a 50% growth cost and
 44 +0.012 for drug-induced tolerance; the schedules with the lowest long-term growth rate are release period 0, dose
 45 2 (growth rate +0.004 per time unit), release period 10, dose 2 (growth rate -0.016 per time unit) and release
 46 period 10, dose 1 (growth rate +0.009 per time unit) respectively. For pre-existing tolerance without a cost, every
 47 release period raises the net growth rate; with the fitness cost, release periods of 5 and 10 time units are within
 48 0.003 per time unit of continuous dosing at doses of 1 and 2; for drug-induced tolerance, dose 1 gives a lower net
 49 growth rate than dose 2 at every release period, and release periods of 20 time units or more raise the growth rate
 50 in every model. All values are single realisations with 300 founder cells.

51 **Supplementary Note 5: Benchmark details**

52 Network inference methods: absolute Pearson and Spearman correlations of $\log(1 + x)$ (Pearson) or ranks
 53 (Spearman); GENIE3 as random-forest importances (200 trees, `sqrt` features) fitted on $\log(1 + x)$ with scikit-learn.
 54 Imputation: kNN-smoothing (one step, $k = 15$, 10 principal components of the Freeman-Tukey transformed,
 55 library-size normalised counts) and MAGIC with default parameters on square-root normalised counts. Metrics:
 56 area under the precision-recall and receiver-operating curves against the undirected (correlation) or directed
 57 (GENIE3) ground-truth adjacency. The cell-cycle-regressed dataset is the residual of $\log(1 + c)$ (concentration c)
 58 after ordinary least-squares regression on gene copy number, cell age and age squared, per gene.

59 **Supplementary Note 6: Inference diagnostics**

60 **Summary statistics and distances.** For Fig. 6a,b the distance between a simulated and the observed sample
 61 of counts is their Wasserstein-1 distance², computed as the mean absolute difference between the two empirical
 62 quantile functions on a grid of 201 probabilities, divided by the observed mean; each simulated sample contains
 63 600 cells (a) or the final snapshot of 500 founders grown for 80 time units (b), against 2 000 observed cells. The
 64 distance between two independent samples at the true parameters (the noise floor) is reported in the summary
 65 tables of the repository. For Fig. 6c the summaries of a treated population are the log surviving fraction on a
 66 grid of 25 time points from the start of the drug to 60 time units later (populations that went extinct contribute
 67 a floor of 10^{-3}) and the concordance of death fates between sisters born in the 20 time units before the drug; the
 68 distance is the root-mean-square difference of these summaries. Each generation's tolerance is the median of the
 69 previous generation's accepted distances, particles are perturbed with a Gaussian kernel of variance twice the
 70 weighted variance of the previous generation, and the first generation samples the prior.



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72 **Supplementary Fig. 2 | ABC-SMC diagnostics. a**, Tolerance and **b**, acceptance rate per generation for the
73 three runs of Fig. 6.

74 The run on non-dividing cells (10 generations, 200 particles) reached a final tolerance of 0.148 with an acceptance
75 rate of 1% in the last generation; the division-aware run (10 generations, 120 particles, each particle simulating
76 a population of 250 founders for 80 time units) reached 0.0841 at 1%; the drug-parameter run (5 generations,
77 100 particles, 200 founders each) reached 0.0727 at 8%. Posterior predictive checks were made by simulating the
78 forward model at the posterior median (Fig. 6d for the drug parameters); the posterior median rather than the
79 mean is reported throughout because the posteriors are asymmetric on the log scale.

80 Supplementary Note 7: Runtime

81 Supplementary Table 3 lists the wall-clock times of the population simulations of Fig. 2f (200 steps of $dt = 0.1$ for
82 random telegraph networks of 10, 50 and 100 genes, four threads of an Intel Xeon processor at 2.1 GHz, Julia
83 1.10). Throughput in cell-steps per second falls with the number of reactions because every step of every cell
84 has to sample the events of all channels; the hybrid kernel is 2 to 5 times faster than the direct method at these
85 tolerances, the gap widening with the number of reactions. The single-cell interface used for the comparison
86 with JumpProcesses.jl (Fig. 2e) allocates a workspace per simulation and is slower in serial execution than
87 JumpProcesses.jl on that one-gene model; the population interface reuses one workspace per thread.

88 Supplementary Table 3. Runtime of population simulations.

genes	cells	kernel	threads	seconds (200	
				steps)	cell-steps / s
10	100	DirectSSA	4	0.38	52,398
10	100	HybridSSATau	4	0.11	180,019
10	1000	DirectSSA	4	1.04	193,080
10	1000	HybridSSATau	4	0.58	346,004
10	5000	DirectSSA	4	4.78	209,388

genes	cells	kernel	threads	seconds (200 steps)	cell-steps / s
10	5000	HybridSSATau	4	2.57	388,683
50	1000	DirectSSA	4	9.72	20,581
50	1000	HybridSSATau	4	2.57	77,671
100	1000	DirectSSA	4	31.41	6,367
100	1000	HybridSSATau	4	6.37	31,373

Supplementary Note 8: Laboratory and clinical reference data

Supplementary Table 4. Reference values used for calibration and validation (transcribed from the cited articles; files in `paper/data/`).

Quantity	Value	Source
U2OS cells tracked at cisplatin addition (7 / 10 / 13 μ M)	232 / 240 / 296	¹ , Table 2
of which died within 72 h	30 / 66 / 176	same
of which divided within 72 h	100 / 75 / 67	same
of which survived without dividing	102 / 99 / 53	same
HCT116 cells at drug addition; death fraction	275; 0.64	¹ , Table 1
Plateau of the HCT116 kill curve	after ~100 h	¹ , Fig 1b
Lineage correlations of fate	present for sisters, first and second cousins; absent for third cousins	¹ , Fig 5a
Pre-resistant melanoma cells	1:50 to 1:500 per marker; EGFR-high cells give 7.9 ± 0.9 fold more resistant colonies	³
N15-0385 glioblastoma doubling time	50 h	⁴
Temozolomide elimination half-life	2.1 h	⁵
RTOG 0525 regimens	150-200 mg/m ² days 1-5 vs 75-100 mg/m ² days 1-21 of 28-day cycles; median OS 16.6 vs 14.9 months	⁶
SWOG S1320 regimens	continuous vs 3 weeks off / 5 weeks on after an 8-week lead-in; median PFS 9.0 vs 5.5 months	⁷

Supplementary Note 9: Migration from version 1

The v1 API (`Donne`, `ssa`, `tauleap`, `tauleapswitch`, `adaptive_tauleap`, `exponential_growth`, `NamedTuple` reactions) remains available through a deprecated compatibility layer that converts `NamedTuple` reactions to `ReactionModels` (reactions whose names contain `act`, `inhib`, `comb_a` or `comb_i` become Hill kinetics with `rate`

96 = [k, n, K]; species whose names contain `on/off` become promoter groups) and maps the algorithms onto the
 97 new kernels. Output shapes are preserved, including the leading `:NULL` column. See `docs/src/migration.md`.

98 **Supplementary Note 10: Relation to previous work**

99 The ingredients of this framework have precedents, and several of its results confirm conclusions reached earlier
 100 by other means. The table separates, claim by claim, what was already established from what this work adds.
 101 Prior results the framework reproduces are evidence that it is behaving correctly, not claims of novelty.

Result in this Article	Already established	What this work adds
Stochastic reaction kinetics inside growing, dividing cells (Fig. 3)	Simulators of stochastic expression with growth and division exist ^{8,9} , including version 1 of this framework ¹⁰	Gene replication, promoter inheritance, a drug layer, lineage statistics, observation models and inference in one forward model, so that a resistance phenotype is a heritable expression state with measurable kinetics rather than an assumed compartment
Simulation of division trees for lineage-tracing methods	TedSim couples expression to division history ¹¹ ; Cassiopeia simulates topologies, heritable fitness and CRISPR barcodes ¹²	Molecular content on the same tree: partitioning at division, promoter states, drug-induced death, and per-cell fates with the molecules that caused them
Heritable expression states decide which cells survive a drug (Fig. 4)	Measured directly in barcoded and time-lapse experiments ^{1,3,13,14}	A generative model in which memory is one promoter timescale, reproducing the reported signatures quantitatively and predicting which of them discriminate pre-existing from induced tolerance
Cell-cycle-aware inference of transcriptional kinetics is necessary (Fig. 3f, Fig. 6b)	Established from data and theory ^{15–17}	A simulator that generates the data such methods assume, and a quantification of the bias incurred by a division-blind fit
Benchmarks of network inference from single-cell data	Extensive benchmarks exist ^{10,18,19}	The growth and cell-cycle confound with known ground truth: how much accuracy is lost in a dividing population, and how much cell-cycle regression recovers (Fig. 5b)

Result in this Article	Already established	What this work adds
The benefit of treatment holidays depends on a cost of resistance and on turnover (Fig. 8a-e)	Compartment models of adaptive therapy ^{20,21}	The same dependence from single cells whose resistance is a heritable expression state; cost and degree of protection become cell properties that lineage experiments measure; partial protection reverses the ranking even without a cost
Perturbing the retention of a resistant state changes the outcome (Fig. 4f)	Phenotypic-switching theory ²² and memory-disrupting compounds ¹³	The timing requirement (disruption must continue during exposure) and a lineage-resolved readout, the number of surviving clones
An intermediate dose can be optimal against drug-induced tolerance (Fig. 4h)	Reported for induced persisters ²³	Reproduced; the accompanying benefit of release periods does not appear here, which the text attributes to the decay of the induced state in this model rather than to a disagreement about data
MGMT expression selects glioblastoma cells under temozolomide (Fig. 8f-h)	Phenotypic selection with stable inheritance ⁴	Pharmacokinetics, the RTOG 0525 regimens, consumption of MGMT by the drug, and the fractionation that minimises the final population at fixed cumulative dose
Dose-dense temozolomide does not improve survival	Clinical result ⁶	A mechanistic account of why, and identification of the rate at which the drug consumes MGMT as the measurement that separates the regimens

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